From 'light crime but severe penalty' to 'punishment fits crime': an empirical study on embezzlement and bribery crime in China

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This paper empirically examines how a leniency legal revision from 'light crime but severe punishment' to 'punishment fits crime'—exemplified by the 2016 new *Judicial Interpretation* of embezzlement and bribery crime in China—impacts the realization of justice and deterrence purpose of criminal law. Employing a novel panel data of embezzlement crimes, bribery crimes, and misappropriation of public funds crimes in 296 Chinese cities from 2001 to 2020 and using the generalized difference-in-differences model to explore the sentence variations of these crimes before and after the *Judicial Interpretation* came into effect, this paper finds that the *Judicial Interpretation* significantly reduces the level of punishment, which is not only because the same crime activity is sentenced lower under the new standard, but also due to the fact that 'punishment fits crime' can prevent offenders from engaging in opportunistic behaviour. In addition, while the *Judicial Interpretation* significantly increases the number of crimes, this does not imply that the efforts toward 'punishment fits crime' condone corruption and do harm to public goods, but achieve substantial improvements in the effectiveness of social governance. The conclusion suggests that the deterrence of criminal law is not necessarily diminished by a relaxation of retributivism, and may provide key evidence for similar criminal law reform of 'punishment fits crime' in many countries.

Keywords: 'punishment fits crime'; embezzlement and bribery crime; generalized difference-in-differences; criminal law reform

I. Introduction

Reform of the criminal law is as old as criminal law itself.[1] Among the many reform opinions, the call for further promotion of 'punishment fits crime' has never ceased since the Age of Enlightenment in the Western world. Theoretically speaking, from the

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[1]See Hans-Heinrich Jescheck, 'The Significance of Comparative Law for Criminal Law Reform' (1981) 5 *Hastings International and Comparative Law Review* 1.

classical school of crime to the positivist school of crime,[2] to many new developments of criminological theories in recent years,[3] the promotion of a 'punishment fits crime' in order to respect and safeguard human rights, effectively punish and prevent crime, and promote the social reintegration of offenders, has become a common value orientation. From the perspective of practice, although social problems such as mass incarceration still exist in some countries,[4] many countries have successfully launched the movement to reform the punishment system of crime, like limiting or even abolishing the death penalty, improving the treatment of prisoners, and wider application of fines,[5] in order to reduce excessively harsh and unjust penalties.

However, China, influenced by a long period of imperial rule and centralized bureaucratic system, has considerable traditions of heavy-penaltyism,[6] where not only the ruler, but also a significant part of the population is often convinced that only heavy penalties can punish offenders and deterring crimes. Into the modern era, the absence of the idea 'punishment fits crime' has led to a series of problems in human rights protection and social governance, calling for the rethinking on these traditions: Should China change the system of heavy-penaltyism and introduce reforms to 'punishment fits crime'? Will such reforms lead to a decrease in the deterrence of criminals, which in

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[2]See George Bryan Vold, *Theoretical Criminology* (Oxford University Press, 1958).

[3]See Marc Ancel, *Social Defence: A Modern Approach to Criminal Problems* (Psychology Press, 1998).

[4]See Bernard Harcourt, 'On the American Paradox of Laissez Faire and Mass Incarceration' (2011-2012) 125 *Harvard Law Review Forum* 54.

[5]See Bingzhi Zhao and Yixiang Jin, 'On the World Background and Chinese Practice of Mitigation of punishment (论刑罚轻缓化的世界背景与中国实践)' (2012) 6 *Journal of Law Application* (《法律适用》) 7.

[6]Ibid, 8; Also see Bingzhi Zhao, *The road of the abolition of the death penalty in China* (中国废止死刑之路探索) (People's Republic Security University Press, 2004).

turn will lead to a series of social problems? These questions have been at the centre of the debate on criminal law reform in recent years in both academic and practical circles.

In the midst of the debate between proponents and opponents of 'punishment fits crime', one of the most well-known and far-reaching legal updates—*Interpretation of Several Issues concerning the Application of Law in the Handling of Criminal Cases of Embezzlement and Bribery*[7] (hereinafter referred to as the *Judicial Interpretation*) promulgated in 2016. The *Judicial Interpretation* greatly raised the monetary amount standard of sentencing for the embezzlement and bribery crime. For example, the sentencing starting point of embezzlement crime and bribery crime[8] increased from the original 5,000 Chinese yuan to 30,000 Chinese yuan, the 'amount is especially large' standard from 100,000 Chinese yuan to 3,000,000 Chinese yuan, and so on. Enjoying the same effect as the law,[9] the implementation of the *Judicial Interpretation* meant that the penalties for embezzlement and bribery crime of the same level would be far less severe than in the past because of the upward revision of the monetary threshold for sentencing.

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[7]Interpretation of Several Issues concerning the Application of Law in the Handling of Criminal Cases of Embezzlement and Bribery (关于办理贪污贿赂刑事案件适用法律若干问题的解释).

[8]Under China's current criminal law system, 'embezzlement and bribery crime' is a category of crimes, including the crime of embezzlement, bribery, misappropriation of public funds, organizational bribery, bribery through the use of influence and advantage and so on. The 'embezzlement crime' and 'bribery crime' mentioned here are specific individual crimes under the whole category 'embezzlement and bribery crime', where the difference in presentation is only for the distinction between the two.

[9]See Shibing Cao, 'The Status of Decisions and Judicial Interpretations of Supreme People's Court (最高人民法院裁判、司法解释的法律地位)' (2006) 3 *China Legal Science* (《中国法学》) 181.

Consequently, this reform is considered a vital effort to shift from 'light crime but severe penalty' to 'punishment fits crime'. Proponents argue that the new *Judicial Interpretation* upwards the monetary standard of sentencing amount, solving the problem of the original standard being too harsh under the continuous economic and social development.[10] Moreover, it places more emphasis on the crime circumstance in sentencing, which is strict but not severe.[11] However, there are also some questionings about the *Judicial Interpretation*. The public doubts whether this relaxation of sentencing standards will condone embezzlement and bribery crime. Compared to robbery, theft and other crimes of illegal possession of the property, the embezzlement and bribery crime sentencing would have been more lenient before the reform, and if the law continues to relax the standards, does this mean that 'We hang the petty thieves and appoint the great ones to public office?'[12]

Although numerous studies have discussed the aforementioned disputes from various viewpoints, the evidence provided based on empirical methods, especially causal inference methods, remains extremely sparse, which leads to the fact that the net effect of the *Judicial Interpretation* has not yet been precisely identified. This paper does not intend to discuss the *Judicial Interpretation* from the view of normative

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[10]See Xianding Pei and others, 'Understanding and Application of the Interpretation of the Supreme People's Court and the Supreme People's Procuratorate on Several Issues concerning the Application of Law in the Handling of Criminal Cases of Embezzlement and Bribery (《关于办理贪污贿赂刑事案件适用法律若干问题的解释》的理解与适用)' (2016) 17 *The People's Judicature* (《人民司法》) 19.

[11]See Xianquan Liu, 'Systematic Review of the Latest Conviction and Sentencing Standards for Embezzlement and Bribery Crimes (贪污贿赂犯罪最新定罪量刑标准体系化评析)' (2016) 5 *Legal Science* (《法学》) 81.

[12]Ibid.

analysis, but instead focuses on the empirical examination of the following two questions: First, in order to discuss whether the *Judicial Interpretation* has facilitated the shift from 'light crime but severe penalty' to 'punishment fits crime', the core is to verify whether the *Judicial Interpretation* significantly reduces punishment to embezzlement and bribery crime offenders, namely, whether successfully safeguard the human rights of offenders—and if it does, what are the mechanisms and reasons for its effect? Second, has the *Judicial Interpretation* led to more cases of embezzlement and bribery crime and a greater negative impact on society, that is, reduced deterrence and prevention of crime? This issue is of great concern to the public and is directly related to the evaluation of the *Judicial Interpretation* and whether similar reforms can be implemented in the future.

In order to study the above two critical issues, this paper obtained the full texts of 67,691 judgments on the embezzlement crimes, bribery crimes and misappropriation of public funds crimes from 2001 to 2020 from the *Chinese Judgement Online* (中国裁判文书网) website, which is the only official website in China that publishes all judicial documents. By matching with macro statistics data from the *China City Statistical Yearbook* (《中国统计年报》), this paper constructs a novel panel data of embezzlement and bribery crime cases for all 296 cities across China. Using the generalized difference-in-differences model, the paper finds that:

First, the *Judicial Interpretation* does reduce the average penalty of embezzlement and bribery crime significantly, which means that the efforts made by this *Judicial Interpretation* to shift from 'light crime but severe penalty' to 'punishment fits crime' are effective. In this paper, a series of robustness tests are conducted to ensure that the results are accurate, and in the analysis of the mechanism, it is found that this reduction is not only because crimes of the same level are sentenced lower under

the new standard, but also because 'punishment fits crime' can prevent offenders from engaging in opportunistic behaviour. More specifically, the reform achieves a more reasonable level of marginal deterrence, and has the function of reducing potential offenders' intensions to commit serious crimes.

Further, this paper also finds that although the *Judicial Interpretation* significantly increases the number of embezzlement and bribery crime, this does not mean that 'punishment fits crime' condones corruption and reduces deterrence. Although the number of cases did rise as a result of *Judicial Interpretation*, the key variable measuring the social harm of crime—the total sentence in each city was significantly lower, and the majority of the additional cases are minor cases, suggesting that in fact the social harm of these crimes is mitigated. The above conclusions clearly indicate the reform value of *Judicial Interpretation*, provide important evidence for the long-standing debate on whether or not to push forward the reform of 'punishment fits crime', and moreover, offer experience and inspiration for the further advancement of similar reforms in the future.

Next, Section II will elaborate on the process of *Judicial Interpretation*, review the previous literature, and propose the paper's research hypotheses; Section III will discuss the core problem: Did the *Judicial Interpretation* reduce the average penalty and protect the human rights? Starting from data acquisition and model construction, empirical results will be given and robustness tests will be conducted, and finally the mechanism of its effect will be investigated; Section IV will focus on a further discussion: Did the *Judicial Interpretation* exacerbate the crime and reduce the deterrence? The discussion will start from the question, provide an answer using the empirical approach, and draw a short summary. Section V concludes.

II. Background, literature, and hypotheses

A. Legislative process

First, it is necessary to shortly review the legislative process of the *Judicial Interpretation*. Prior to the *Judicial Interpretation*, the penalties for embezzlement and bribery crime were mainly stipulated in the *Criminal Law of the People's Republic of China 1997*[13] (hereinafter referred to as the *Criminal Law 1997*). For example, for the embezzlement crime and the bribery crime, individuals who have engaged in graft with an amount of more than 100,000 Chinese yuan are to be sentenced to more than 10 years of fixed-term imprisonment or life imprisonment, 50,000 to 100,000 Chinese yuan corresponds to more than 5 years of imprisonment, and 5,000 to 50,000 Chinese yuan corresponds to imprisonment more than 1 year but less than 7 years.[14]

For the crime of embezzlement and bribery, the monetary amount standard is an important basis for conviction and sentencing. However, along with the continuous development of China's economy and society, the monetary amount standard for the crime remained unchanged until 2016, which means that in 2015, when the per capita GDP was already 7.7 times higher than in 1997 (about 49,922 Chinese yuan) and the per capita consumption was already 6.3 times higher than in 1997 (about 18,857 Chinese yuan),[15] the sentencing threshold for embezzlement crime and bribery crime remained at 5,000 Chinese yuan. As a result, if this set of standards is still adhered to

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[13]Criminal Law of the People's Republic of China 1997 (中华人民共和国刑法 1997).

[14]Ibid, Art 3.

[15]See China National Bureau of Statistics, 'National Data' <https://data.stats.gov.cn/> accessed 1 Feb 2024.

without change, the punishment for embezzlement and bribery crime may be too harsh and become 'light crime but severe penalty'.[16]

Therefore, the *Amendment (IX) to the Criminal Law 1997*,[17] (hereinafter referred to as the *Amendment (IX)*) made a significant amendment to this problem, eliminating the provision of specific monetary amounts and replacing it with vague standards such as 'the amount is relatively large', 'huge', 'especially huge', and so on.[18] After the *Amendment (IX)* came into effect, in order to further clarify the application of the law, the *Judicial Interpretation* issued in 2016 further clarified the monetary amounts corresponding to the above-mentioned vague standards.[19] Here, the paper presents graph of the changes in the penalty standards for the three crimes of embezzlement, bribery and misappropriation of public funds[20] in Figure 1-2. For detailed tables of specific changes about monetary amount standards, please see Appendix I.

Figure 1. Changes of the penalty standards for embezzlement crime and bribery crime before and after the *Judicial Interpretation*[21]

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[16]See Bingzhi Zhao, 'On the Improvement of the Criminal Rule of Law against Corruption in China (论我国反腐败刑事法治的完善)' (2013) 3 *Contemporary Law Review* (《当代法学》) 54.

[17]Amendment (IX) to the Criminal Law 1997 (刑法修正案（九）).

[18]Ibid, Art 44.

[19]*Judicial Interpretation* (n 7) Art 1-11.

[20]Because the provisions before and after the *Judicial Interpretation* do not correspond exactly, the multiplicity of change is calculated by linear method, and robustness test will be given later.

[21]As mentioned above, the increase in the sentencing monetary amount standards means that the penalties for the same level's crime decrease. To avoid confusion regarding the 'increase' and 'decrease' in this paper, the y-axis in the Figure 1-2 represents 'Penalty for every 100,000 Chinese yuan embezzled or bribed (measured in months)', which clearly illustrates the significant reduction in penalties due to the *Judicial Interpretation*.

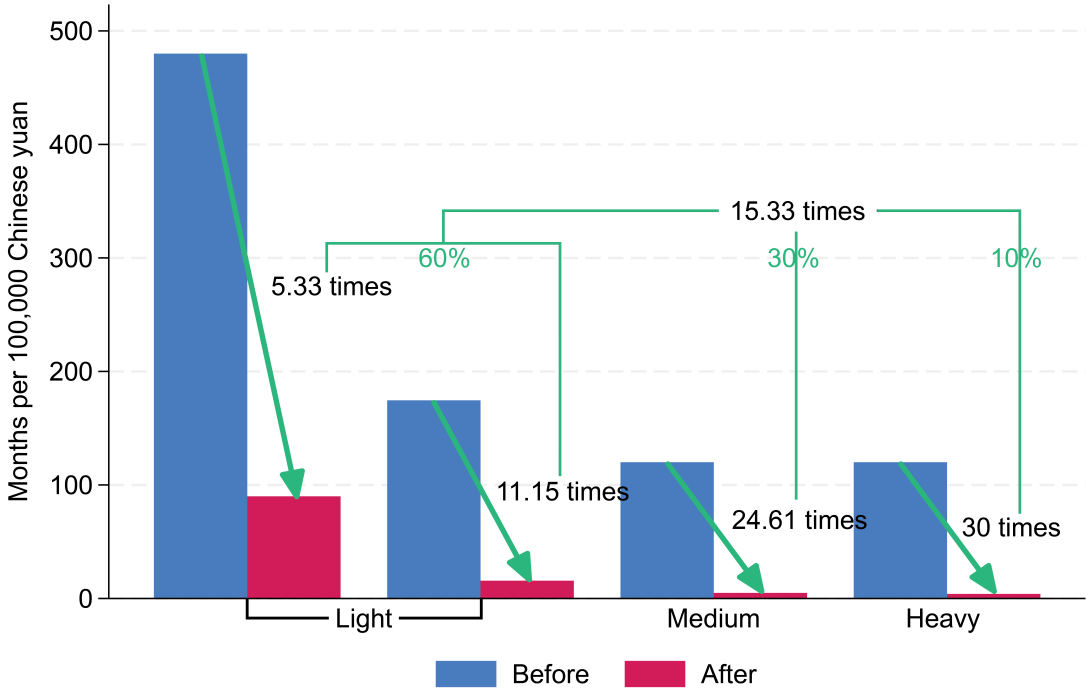
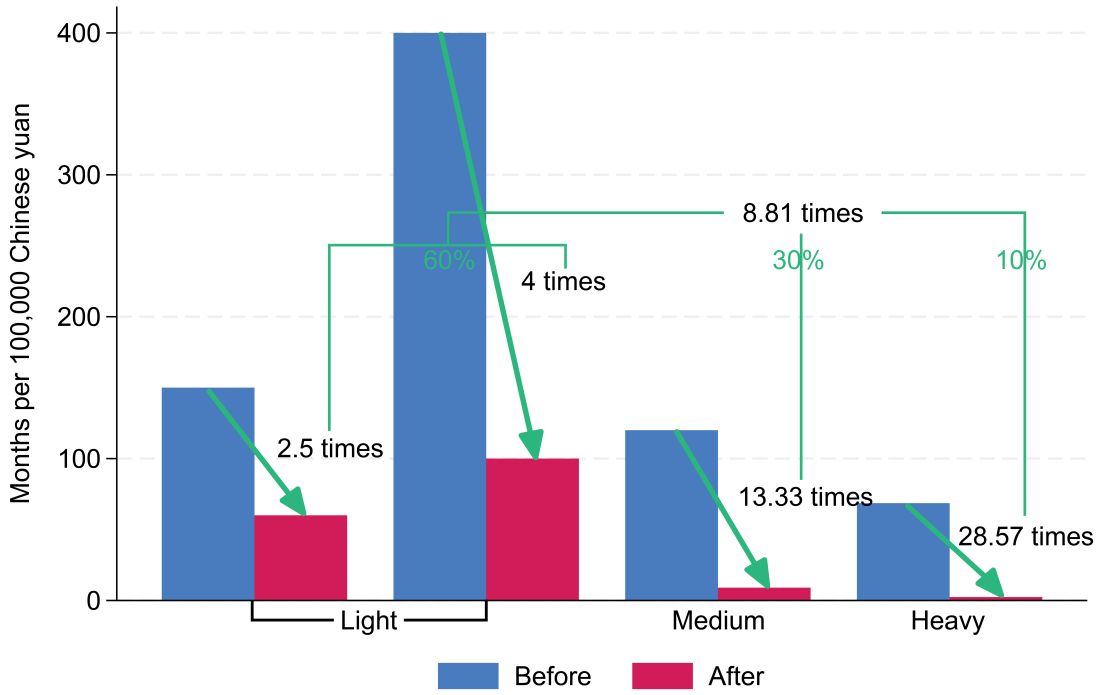


Figure 2. Changes of the penalty standards for misappropriation of public funds crime before and after the *Judicial Interpretation*



From Figure 1-2, it is clear that the penalty standards for embezzlement and bribery crime have dropped significantly following the introduction of the *Judicial*

Interpretation. In terms of its subdivision, the decrease in the embezzlement crime and bribery crime was much greater than that of misappropriation of public funds crime, which will be the cornerstone of the later empirical study.

B. Literature review

'Punishment fits crime', as known as 'balance of punishment and crime', 'crime equalling punishment', or 'proportion between punishment and crime', means that not only does the penalty need to be commensurate with the egregiousness of the crime, but also the manner in which it is applied should also correspond to the seriousness of the offense.[22] The reason behind is simple, that 'if an equal penalty is attached to two crimes of unequal injury to society, the greater crime of the two, if it promise a greater advantage than the other, will have no stronger motive in restraint of its perpetration.'[23] On this basis, Bentham further proposed five principles for 'punishment fits crime' based on utilitarianism,[24] Kant and others, based on retributivism, similarly believe that crime and punishment should achieve a balance to equalize the scales of justice.[25] All of these form the ideological basis of the time-honored 'punishment fits crime'.

In modern times, as Durkheim said, penalties should necessarily be progressively reduced in line with industrialization and the social division of labour, and modernization meant, in a sense, lighter penalties.[26] A group of modern criminal law theories, represented by the 'new social defence theory', advocates that the educational function of punishment should be prioritized over punishment and retribution,

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[22]See Cesare Beccaria, *On Crimes and Punishments* (Cambridge University Press, 1995).
[23]Ibid.
[24]See Jeremy Bentham, *Theory of Legislation* (Harcourt Brace and Company, 1931).
[25]See Immanuel Kant, *Critique of Practical Reason* (Cambridge University Press, 2015).
[26]See Steven Lukes and Andrew Scull, *Durkheim and the Law* (Bloomsbury Publishing, 2017).

emphasize the protection of the rights and dignity of criminals through 'punishment fits crime', and aim to help offenders in reintegrating into society.[27] In conclusion, the 'punishment fits crime' as a part of legal justice is a consensus that has basically gained wide recognition.

However, the above consensus is not quite the same in China, especially before the Reform and Opening Up (改革开放) in 1978, from which China embarked a new road to 'socialism with Chinese characteristics (中国特色社会主义)'.[28] For a long time, China has in fact adhered to a tradition of heavy criminalization and has developed a certain culture of heavy-penaltyism from state to people.[29] In the case of embezzlement and bribery crime, for example, the criminal law of Qin Dynasty stipulates that 'even if a bribe of only one coin is taken, the person shall be branded on the forehead and punished to building the city.'[30] In Ming Dynasty, embezzlement and bribery crime offenders were sanctioned with torture, such as amputation, flaying or even slow slicing by the special law called *Ming Da Gao*.[31] Into modern times, although heavy-

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[27]See Marc Ancel (n 3).

[28]See Ezra Vogel, *Deng Xiaoping and the Transformation of China* (Harvard University Press 2011).

[29]See Bingzhi Zhao (n 6).

[30]See Collation Team of Shuihudi Bamboo Strips of the Qin Dynasty, *Shuihudi Bamboo Strips of the Qin Dynasty* (《睡虎地秦墓竹簡》) (Cultural Relics Press, 1990). The original text here is '通一錢黥城旦罪', in which 'one coin' ('一錢') is made from roughly 10 grams of copper and represents its equivalent value.

[31]See Yifan Yang, *Ming Da Gao Studies* (《明大誥研究》) (Jiangsu People's Publishing House, 1988).

penaltyism has been corrected to a certain extent,[32] China's punishment to crime is still relatively heavy compared to other countries all over the world.[33] In particular, in order to quell the destruction of the rule of law caused by the Cultural Revolution (文化大革命), China has launched several 'strike hard (严打)' movements to severely punish criminal offenses since 1983,[34] but it has proved to be less effective than desired.[35] Along with the introduction of the Western theory of 'punishment fits crime' to China, some scholars have explicitly supported similar reforms in China to modernize and legalize criminal law and criminal-social governance; However, a significant number of scholars and practitioners still advocate heavy-penaltyism, in fear of 'exonerating and indulging criminals'.[36] Specifically on the issue of embezzlement and bribery crime, the differences between the two parties are even more serious:

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[32]See Yunsheng Chen, *Against torture—Legal policy and human rights protection in today's China* (反酷刑——当代中国的法治和人权保护) (Social Science Academic Press of China, 2000).

[33]See Børge Bakken, 'China, a Punitive Society' (2011) 6 *Asian Criminology* 33; Also see Ronggong He, 'Critique of Philosophy of Law on the "Excessive Use of Criminal Law" in Social Governance (社会治理"过度刑法化"的法哲学批判)' (2015) 27 *Peking University Law Journal* (《中外法学》) 527.

[34]See Harold Tanner, *Strike Hard! Anti-Crime Campaigns and Chinese Criminal Justice, 1979-1985* (Cornell University Press, 1999); Also see Bin Liang, 'Severe strike campaign in transitional China' (2005) 33 *Journal of Criminal Justice* 387.

[35]See Xingliang Chen, 'Study on the Criminal Policy of Combining Punishment with Leniency (宽严相济刑事政策研究)' (2006) 1 *Legal Science Magazine* (《法学杂志》) 18; Also see Susan Trevaskes, *Courts and Criminal Justice in Contemporary China* (Lexington Books, 2007).

[36]For instance, among a large number of regulations on intra-CCP governance of corruption, such as the *Opinions on Strengthening the Construction of a Culture of Integrity in the New Era* (《关于加强新时代廉洁文化建设的意见》) and other regulations, '猛药去

Most scholars have expressed approval and praise of the *Judicial Interpretation*, whose views are: First, the *Judicial Interpretation* is not the first relaxation of the embezzlement and bribery crime and there are precedents before,[37] which is to ensure that the penalties are always equal to the social harm of crimes, and previous reforms have not created particularly serious problems.[38] Since GDP per capita has increased more than 6 or 7 times over these years, raising the sentencing standards too little would not be effective.[39] Second, scholars who advocate punishment fits crime' believe that the lenient sentences have their justification as an instrument of criminal policy,[40] claiming that the combination of leniency and severity is a concrete expression of both the opposition and unity of retribution and utilitarianism, rationalism and modesty, and punishment and prevention.[41] The combination of leniency and severity is formulated

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疴、重典治乱' is always a popular expression, which means 'we must use strong medicine to remove illness and heavy discipline to control chaos'.

[37]See Xianding Pei and others (n 10).

[38]See Qinzhong Lu, 'On Legislative Amelioration of the Penalty of Bribery Crime (我国受贿罪刑罚的立法完善)' (2008) 3 Journal of National Presecutors College (国家检察官学院学报) 83.

[39]See Bingzhi Zhao, 'A Brief Discussion of the Latest *Judicial Interpretation* of the Crime of E mbezzlement and Bribery Conviction and Sentencing Standards (略谈最新司法解释中贪污受贿犯罪的定罪量刑标准)' (18 April 2016) <https://www.chinacourt.org/article/detail/2016/04/id/1841810.shtml> accessed 15 January 2024.

[40]See Guoxiang Sun, 'Gains and Losses of Amendment to Criminal Law on Embezzlement and Bribery Crimes (贪污贿赂犯罪刑法修正的得与失)' (2016) 3 *Journal of Southeast University (Philosophy and Social Science)* (《东南大学学报（哲学社会科学版）》) 73.

[41]See Wei Jiang and Yurong Lu, 'The Dialectical Relationship Between Leniency and Severity in Criminal Policy (宽严相济刑事政策的辩证关系)' (2007) 6 *Criminal Science* (《中国刑事法杂志》) 11.

according to the different circumstances of the criminal offenders, and is of great significance to educate and rehabilitate of most offenders, isolate and combat the minority, and divide and disintegrate the 'enemy class'.[42] At the same time, it also serves to protect human rights and reflects the principle of equity.[43] Finally, many studies based on judicial cases and media reports point out that the sentencing amount standard in the *Criminal Law 1997* was not fully implemented before 2016.[44] For example, some courts have substantially raised the standard of filing cases internally, from generally not pursuing embezzlement and bribery crimes below 10,000 Chinese yuan to the current rule of 50,000 Chinese yuan.[45] A large number of suspended sentences, and minor sentences are also being applied by the judiciary.[46] Therefore, the *Judicial Interpretation* is an adjustment in line with the reality.

At the same time, there is some opposition to the *Judicial Interpretation* from the academic community, but it is not mainly about whether the sentencing standards should be revised, but about the magnitude of the revision since it is believed that the current revision is too large and may condone the crime. Some scholars have pointed

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[42]See Mingxuan Gao, *The Conception and Brith of the Criminal Law of the People's Republic of China* (《中华人民共和国刑法的孕育与诞生和发展完善》) (Peking University Press, 2012).

[43]See Wei Jiang and Yurong Lu (n 41) 16.

[44]See Mingkai Zhang, 'Judicial and Legislative Development Direction of the Crime of Embezzlement and Bribery (贪污贿赂罪的立法与司法发展方向)' (2017) 1 *Tribune of Political Science and Law* (《政法论坛》) 5.

[45]See Guoxiang Sun, 'The Disorder of Sentencing Imbalance in Corruption and Bribery Crimes and its Relief (宽严皆失：贪污贿赂犯罪的量刑失衡之乱象及纾解)' (2009) 9 *Journal of Gansu Institute of Political Science and Law* (《甘肃政法学院学报》) 27.

[46]See Xuanguo Xiong, *Reference of Criminal Trial (Ep. 57)* (《刑事审判参考（第 57 卷）》) (Law Press China, 2007).

out that breaking through the dilemma of rule of law practice does not lie in the frequent formulation of judicial interpretation, but in the practical improvement of legislative techniques.[47] It is argued that the introduction of the new *Judicial Interpretation* will bring about an increase in the number of corruption crimes occurring, and that although the sentence is nominally strict, it actually relaxes the penalties for the perpetrators.[48] Also, the *Judicial Interpretation* may trigger the contradiction of unbalanced conviction and sentencing standards within the property crime system.[49] Some scholars have also questioned whether there is any inherent scientific support for keeping the multiples of the increase in sentencing standards in line with the multiples of GDP growth.[50] Some other scholars raised doubts in response to the specific amount of the division of the standard change, such as that the standard of 100,000 Chinese yuan corresponding to more than ten years of fixed-term imprisonment or life imprisonment is too high, and so on.[51] In general, although the opponents' challenges vary, rather than opposing the *Judicial Interpretation*, they are proposing amendments to the specific provisions of the *Judicial Interpretation*.

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[47]See Xianquan Liu (n 11) 79. The rationale behind this can actually be traced back to Becker's analysis of the criminal behavior from law and economics, see Gary Becker, 'Crime and Punishment: An Economic Approach' (1968) 76 *Journal of Politicial Economy* 169.

[48]See Guoxiang Sun (n 40) 72.

[49]See Xianquan Liu (n 11) 84.

[50]See Jie Xie, 'System Optimization and Rule Supplementation in the Governance of Corruption and Bribery Crimes—Based on the Dual-Oriented Reflection on the Latest Judicial Interpretation of Law and Economy (贪污贿赂犯罪治理的制度优化与规则补充——基于对最新司法解释的法律与经济双面向反思)' (2016) 6 *Political Science and Law* (《政治与法律》) 32.

[51]See Jinting Deng, 'Empirical Study of the Crime Equalling Punishment in Bribery Crime (受贿罪罪刑均衡实证研究)' (2022) 2 *The Jurist* (《法学家》) 104.

By reviewing the different views of the two major schools of thought mentioned above, a core of the discussion comes to the fore: Whether praising the *Judicial Interpretation* or disapproving of its specific provisions, the focus of their attention is the effect and utility of the *Judicial Interpretation*, specifically its utility of promoting the balance of crime and penalty, reducing the occurrence of crime, and combining the leniency and severity. Thus, did the *Judicial Interpretation* actually reduce the average sentence? Furthermore, did it increase the number of cases? What are the reasons and mechanisms by which they work? Unfortunately, the few existing empirical studies mainly focus on the impact of various factors and circumstances within the crime on sentencing,[52] or descriptively present the distribution and proportion of sentencing,[53] but fail to provide an answer to the above cause and effect. Therefore, the marginal contribution of this paper will help fill in the gaps in the above questions.

C. Research hypotheses

Did the *Judicial Interpretation* reduce the average penalty, that is, protect the human right of offenders? Many people may think that the answer to this question is undeniable: Since the sentencing amount standards have been significantly increased, the perpetrators of the same amount of embezzlement and bribery crime will receive far lower penalties after the introduction of the *Judicial Interpretation*, and therefore the *Judicial Interpretation* should have lowered the average penalty. However, as reviewed

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[52]See Yong Guo and Songfen Li, 'An Empirical Study of Criminal Sentencing in Corruption and Bribery Cases (贪污受贿案件刑事量刑的实证研究)' (2018) 1 *Comparative Economic and Social Systems* (《经济社会体制比较》) 77.

[53]See Yuncang Song, 'Research on the Balance of Sentencing in Corruption and Bribery Cases (贪污受贿案件量刑均衡问题研究)' (2006) 2 *Criminal Law Review* (《刑事法评论》) 365.

in the literature above, such a conclusion may be hasty: First, there were many precedents of lenient sentencing in practice even before the *Judicial Interpretation* was introduced, and the provisions in the *Criminal Law 1997* have been factually broken. Therefore, if the *Judicial Interpretation* is merely an ex post facto retrospective recognition of changes in adjudicative norms, it may not serve to reduce sentences. Second, without empirical research to identify the net impact of the introduction of the *Judicial Interpretation*, it is difficult to say whether the change in average sentence length is the result of the *Judicial Interpretation*, because many factors such as economic development, industrial structure, population, and fiscal revenue may influence the occurrence of embezzlement and bribery crime. Finally, as also mentioned above, some scholars questioned the adjustment of sentencing amount standards under the new *Judicial Interpretation*, and it is not entirely unlikely that the sentence will rise rather than fall due to unreasonable adjustments. Therefore, the paper proposes a set of hypotheses which are opposite to each other:

Hypothesis 1a: The *Judicial Interpretation* did reduce the average penalty.

Hypothesis 1b: The *Judicial Interpretation* did not reduce the average penalty.

At the same time, as many scholars and people fear, if the above question is confirmed, that is, the introduction of the *Judicial Interpretation* has indeed lowered the average sentence, which may in turn means that the overall deterrence on embezzlement and bribery crime has decreased, will this lead to a rise in the number of crimes? If so, how is this increase viewed and how can we respond to it? This question will likewise serve as a further research question for this paper. Therefore, the paper goes on to propose two opposing hypotheses, which are the second set of hypotheses:

Hypothesis 2a: The *Judicial Interpretation* did exacerbate the crime.

Hypothesis 2b: The *Judicial Interpretation* did not exacerbate the crime.

III. Did the *Judicial Interpretation* reduce the average penalty?

A. Methodology and data

As the beginning of the empirical section, this subsection focuses on the empirical models and data applied throughout the paper. With the 'credibility revolution' in empirical research, the traditional econometric methods represented by simple OLS regression have fallen behind because most of time they can only study correlation but not causation. A series of methods based on the potential outcome model, such as matching, instrumental variable, difference-in-differences, and regression discontinuity, have become the latest research paradigm in empirical research.[54] Among them, the difference-in-differences method (hereinafter referred to as DiD) is the research method that will be used in this paper.

The DiD model is an excellent tool for measuring the impact of a policy change and is based on the following simple idea:[55] Assume a policy whose occurrence divides the continuous time into two periods, before and after the occurrence of the policy. Before the policy occurs, all individuals are unaffected, while after the policy occurs, a portion of individuals, that is, the treatment group, are affected by the policy, while the rest of individuals, that is, the control group, are not affected by it. In this way, if individuals in the treatment and control groups have extremely similar trends before the occurrence of the policy, but show a gap immediately after the occurrence of the policy,

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[54]See Guido Imbens and Donald Rubin, *Causal Inference for Statistics, Social and Biomedical Sciences* (Cambridge University Press, 2015); Also see Wei Huang and others, 'From Difference-in-Differences to Event Study (从双重差分法到事件研究法)' (2022) 2 *Industrial Economics Review* (《产业经济评论》) 17.

[55]See Michael Lechner, 'The Estimation of Causal Effects by Difference-in-Difference Methods' (2011) 4 *Foundations and Trends in Econometrics* 165.

it is reasonable to assume that the occurrence of the policy causes the gap. If the above process is expressed mathematically, there will be the following equation:

$$Y_{it} = \alpha + \beta D_i + \gamma T_t + \delta (D_i \times T_t) + \sum \eta Z_{it} + \varepsilon_{it} \tag{1}$$

in which Y_{it} is the result variable, D_i means whether the individual i is affected by the policy (if the individual i is affected, than $D_i = 1$, else $D_i = 0$), T_t represents whether the policy has occurred at time t (if the policy has occurred, $T_t = 1$, else $T_t = 0$), and $D_i \times T_t$ is the product of D_i and T_t. α, β, and γ is the coefficient of D_i, T_t, and $D_i \times T_t$. Z_{it} is a set of control variables and ε_{it} is the residuals. So the coefficient δ can measure the policy's impact, which is because:

Table 1. The reason why the coefficient δ represents the policy's impact

$\mathrm{E}(Y_{it} \mid D_i,\ T_t)$	$T_t = 0$	$T_t = 1$	Difference
$D_i = 0$	α	$\alpha + \gamma$	γ
$D_i = 1$	$\alpha + \beta$	$\alpha + \beta + \gamma + \delta$	$\gamma + \delta$
Difference	β	$\beta + \delta$	δ

However, in the specific practice of this paper, the standard DiD described above is difficult to apply, which is because the *Judicial Interpretation* introduced in 2016 impacts almost all the crimes in the embezzlement and bribery crime, and almost all the crimes in the category, the research object of this paper, are included in the treatment group, in which case the control group is missing. If we choose other crimes with the purpose of unlawful appropriation of property, such as robbery, as a control group, it is again difficult to ensure that and embezzlement and bribery crime have extremely similar trends before the policy occurs. Fortunately, the 2016 *Judicial Interpretation* raises the sentencing standard differently for different crimes in

embezzlement and bribery crime (just as shown in Figure 1-2), allowing us the flexibility to use the DiD method to construct the following equation:

$$Y_{it} = \alpha + \beta I_i + \gamma T_t + \delta (I_i \times T_t) + \sum \eta Z_{it} + \varepsilon_{it} \tag{2}$$

this method, known as the generalized DiD, was first applied by Nathan Nunn and Nancy Qian to study the effect of the spread of potato cultivation on population growth in Europe.[56] Here, D_i is replaced by I_i, which means the intensity of the impact of the policy on individual i. In Figure 1-2, the paper obtained different sentencing increases by comparing the legal texts for embezzlement crime and bribery crime, as compared to misappropriation of public funds crime, which is exactly the value corresponding to I_i here. What remains constant is that the coefficient δ of the cross-product term remains the policy effect we are interested in.

The data for this paper was obtained from the *China Judgement Online* website, which is the official Chinese website for publicizing the vast majority of judgments. In this paper, we compiled the nationwide verdicts on embezzlement crimes, bribery crimes and misappropriation of public funds crimes from 2001 to 2020,[57] and obtained 67,691 full-text verdicts after removing duplicate samples, eliminating irrelevant

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[56]See Nathan Nunn & Nancy Qian, 'The Potato's Contribution to Population and Urbanization: Evidence from a Historical Experiment' (2011) 2 *The Quarterly Journal of Economics* 593.

[57]There are two main reasons why the time window ends in 2020 and does not continue to today: First, the policy effect usually does not last too long, and the empirical results may not be accurate if too many post-2016 cases are included; Second, and most importantly, fewer and fewer cases are being uploaded after 2020 because of the rise of web crawler technology, the protection of personal privacy, and the control of social opinion, so that including these cases may lead to biases in econometric estimates.

verdicts, and sifting out non-public cases. Since this paper focuses not on individual cases but on crimes within the society as a whole, this paper also matches these 67,691 data with data from the *China City Statistical Yearbook* according to the cities where the judgment courts are located, and obtains panel data in 296 cities (292 prefecture-level cities and four direct-administered municipalities Beijing, Tianjin, Shanghai and Chongqing) from 2001 to 2020, including average sentence, number of cases, total sentence, and control variables such as population, GDP, GDP growth rate, share of secondary industry, share of tertiary industry, fiscal revenue, social consumer goods, road area, and road passenger volume. Descriptive statistics of the data are shown in the following table:

Table 2. Descriptive statistics

Variable	N	Mean	Median	SD	Min	Max
Average sentence	7,159	35.14	27.60	29.98	0.00	350.00
Number of cases	7,159	9.46	6.00	11.77	1.00	179.00
Total sentence	7,159	309.91	160.01	506.05	0.00	10,168.19
T_t	7,159	0.55	1.00	0.50	0.00	1.00
I_i	7,159	12.72	14.34	2.52	8.81	14.34
$I_i \times T_t$	7,159	7.02	8.81	6.57	0.00	14.34
Log of population	7,159	5.96	6.00	0.70	2.94	8.14
Log of GDP	7,159	7.35	7.29	1.02	3.46	10.56
GDP growth rate	7,159	7.84	8.00	3.93	-20.63	28.60
Share of secondary industry	7,159	45.17	45.75	10.62	10.60	81.82
Share of tertiary industry	7,159	43.39	42.68	10.39	15.16	83.87
Log of fiscal revenue	7,159	14.04	13.93	1.11	9.40	18.09
Log of social	7,159	15.68	15.64	1.05	11.56	18.89

consumer goods						
Log of road area	7,159	7.19	7.09	1.00	3.95	10.34
Log of road passenger volume	7,159	8.29	8.30	1.12	2.20	12.18

A few points to note about the data are: First, all sentence length data are in months, and in processing the data, this paper follows the established literature by discounting suspended sentences and setting life sentences at 300 months and death sentences at 350 months.[58] Secondly, to better fit the linear trend, this paper takes logarithms for variables such as population and GDP. Finally, because there are missing data in the data source, for these missing values, linear interpolation is performed in this paper. The amount of interpolated data is minimal compared to the total so it will not affect the empirical results.

B. Empirical result

Having presented the methodology and data, the first question to be examined is whether the *Judicial Interpretation* has reduced the average sentence for embezzlement and bribery crime. Therefore, with the average sentence as the dependent variable and T_t, I_i, and their cross product term $I_i \times T_t$ as the independent variable, a generalized DiD regression can be constructed as follows:

$$Average\ sentence_{it} = \alpha + \beta I_i + \gamma T_t + \delta (I_i \times T_t) + \varepsilon_{it} \quad (3)$$

this most basic regression is shown in the Regression (1) of results in Table 3. On this basis, the paper add the two-way fixed effects commonly used for panel data, controlling for city and year variables, and started using clustering standard error by

[58]See Jinting Deng (n 51) 105.

cities:

$$Average\ sentence_{it} = \alpha + \sum \beta u_i + \sum \gamma v_t + \delta (I_i \times T_t) + \varepsilon_{it} \quad (4)$$

whose results are shown in the Regreesion (2) of Table 3. On this basis, all control variables are then added to the regression:

$$Average\ sentence_{it} = \alpha + \sum \beta u_i + \sum \gamma v_t + \delta (I_i \times T_t) + \sum \eta Z_{it} + \varepsilon_{it} \quad (5)$$

to obtain the results in the third column Regreesion (3).

Table 3. Empirical result

Variables	(1) Average sentence	(2) Average sentence	(3) Average sentence
$I_i \times T_t$	-1.087*** (-5.02)	-1.061*** (-4.14)	-1.035*** (-4.11)
Control variables	No	No	Yes
City FE	No	Yes	Yes
Year FE	No	Yes	Yes
Observations	7,159.00	7,159.00	7,159.00
Number of cities	296.00	296.00	296.00
R-squared	0.103	0.148	0.154

Note: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

It can be found that the coefficients δ we focus on, representing the impact of the policy, are always significant and their values do not change much, whose values are negative, so Hypothesis 1a was confirmed to be valid, reviewing the first set of hypotheses of the study: The *Judicial Interpretation* did significantly reduce the average

sentence of offenders, which is one of the purposes of its introduction—to promote the balance of crime and penalty, and protect the human right of offenders.

Such a conclusion is helpful in answering the many queries mentioned above. The significance of the empirical results implies that, although there was a tendency to leniently apply penalties in judicial practice before the introduction of the *Judicial Interpretation*, the effect of *Judicial Interpretation* was never 'retroactive', but had the effect of significantly promoting the 'punishment fits crime' and further reducing penalties. In addition, the empirical results presented in this paper are net, excluding the influence of economic development, industrial changes, population and other factors. Finally, the decrease in average penalties also partially proves that there is no major failure in the formulation of *Judicial Interpretation* that increases penalties instead.

C. Robustness test

The above regressions are simple and clear, but a series of robustness tests are also done in this paper to prove the reliability of the conclusions. The first is the parallel trend test. As presented above, in the DiD model, the treatment and control groups should follow some common trends before the policy occurs to study the impact of the policy.

Figure 3. Parallel trend test (by line graph)

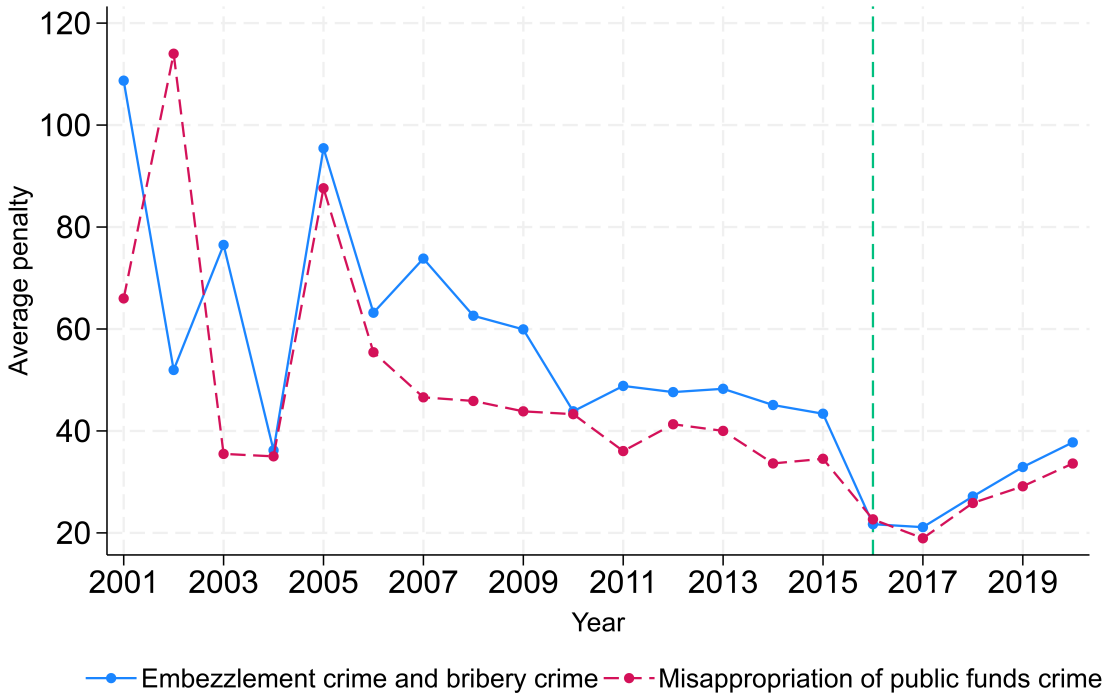

Figure 4. Parallel trend test (by regression including lag and lead terms)

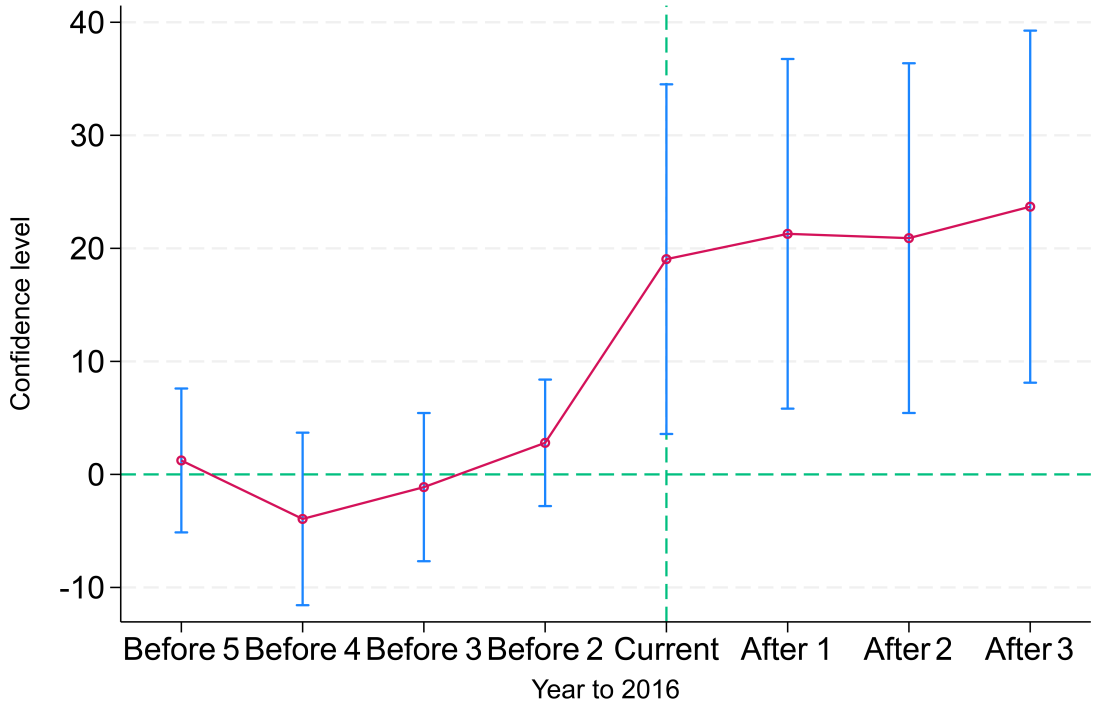

Figure 3 presents the line graphs of the treatment and control groups. Before 2006, the mean values of the two fluctuated due to the small number of samples, but afterwards the trend of the two was very similar and they showed a significant decrease

in 2016. Figure 4 is a more precise illustration using the regression including lag and lead terms.[59] As can be seen in the figure, the confidence intervals of the coefficients for all previous years contain zeros and the coefficients are not significant, but since 2016, the situation immediately shifts and the zeros are outside the confidence intervals and the coefficients begin to be significant, indicating that a parallel trend is satisfied and that the treatment effect of *Judicial Interpretation* does exist.

On this basis, we also need to revisit the above empirical results, as many areas remain questionable. First, the changes in sentencing amount standards before and after the introduction of the *Judicial Interpretation* shown in Figure 1-2 are not completely accurate calculations, and the laws before and after cannot correspond exactly so there may be errors. This paper finds that no matter how to divide the categories and calculate the multipliers, the increases in standards for embezzlement crime and bribery crime are always higher than that for misappropriation of public funds crime, so it is worth considering the sample of embezzlement crime and bribery crime corresponds to 1 and misappropriation crime corresponds to 0, to use the standard DiD to test the robustness. As shown in the first column Regreesion (4) of Table 4, the results are still significant, indicating that the calculation of this partial multiplier does not affect the correctness of the results.

Table 4. Robustness test

Variables	(4) Average sentence	(5) Average sentence	(6) Average sentence	(7) Average sentence	(8) Average sentence
$D_i \times T_t$	-6.747***				

[59] To avoid multicollinearity, the year before the shock time point (Before 1) is not included in the regression.

	(-4.11)				
$I_i \times T_t$		-1.073***	-1.046***	-0.972***	-0.433
		(-4.11)	(-4.12)	(-3.84)	(-0.83)
Control variables	Yes	Yes	Yes	Yes	Yes
City FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Observations	7,159	7,159	7,159	7,024	7,159
Number of cities	296	296	296	292	296
R-squared	0.154	0.153	0.153	0.159	0.151

Note: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Second, in the data treatment, this paper sets the life sentence and death sentence reprieve to 300 and 350 months, and although the literature supports such an operation, it cannot be completely accurate. Therefore, this paper tries again to remove these variables for regression, and the results are shown in the second column Regreesion (5) of Table 4, which are still significant. Similarly, in the face of possible challenges regarding control variables, the paper randomly excludes half of the control variables and the regression results remain significant, which is shown in the Regreesion (6). Also, municipalities directly under the central government are on the same level as provinces in China, will the inclusion of the four cities of Beijing, Tianjin, Shanghai and Chongqing in the regression cause bias? To verify this, the Regreesion (7) of the table presents the results with the four cities removed, and the coefficients are still significant.

Finally, it is necessary to conduct the placebo test, that is, to change the time point of the shock and observe whether the regression remains significant. As the public knows, since the 18th National Congress of the Communist Party of Chinain 2013, anti-

corruption efforts have been higher, could it be that this high-pressure stance has deterred crime and thus led to a decline in sentences? The political fight against corruption has been carried out on a large scale since 2013, and if the change in average sentence length is responsible for it, the results should also be significant using 2013 as the year of policy occurrence for the DiD model in this paper. To verify this, this paper again moves the policy occurrence time point from 2016 to 2013, and the results are shown in the Regreesion (8) in the fifth column of the table, where the coefficients are not only insignificant but also very far from significant results, indicating that the impact comes precisely from the *Judicial Interpretation* in 2016. In addition, the policy role of the anti-corruption fight is consistent for embezzlement crime and bribery crime, as well as misappropriation crime, and is part of term ε_{it} in the model above, which has been subtracted after difference-in-differences.

D. Mechanism analysis

After the robustness test above, the findings in the empirical results subsection are proven: Hypothesis 1a is valid, and the *Judicial Interpretation* does reduce the average sentence. On this basis, what causes this effect?

One explanation is quite obvious: the introduction of *Judicial Interpretation* allows the same level of crimes to be sentenced lower than before under the new standards, thus lowering the average sentence. One of the goals of the *Judicial Interpretation* is to promote the shift from 'light crime but severe penalty' to 'punishment fits crime', so such an explanation is consistent with the purpose of it. However, it may be disappointing if only such an impact mechanism exists. Such an explanation implicitly assumes the *Judicial Interpretation* does not in any way change the occurrence of crimes before and after its introduction, but merely confers different

penalties for the same level of crimes. Although a lot of established literature explores the rationality of this *Judicial Interpretation* from such an assumption, this paper argues that both theoretically and empirically, it can confirm that there is room for improvement of the above theory: There is by no means just one mechanism for the *Judicial Interpretation* to work.

Theoretically, this issue essentially comes back to rationales of criminal law. The mainstream view is that the functions of criminal law should include deterrence function, retribution function and human right protection function.[60] This means that criminal law not only needs to prevent and deter the occurrence of crime, but also needs to punish crime, while protecting the basic rights and interests of the offender. From the perspective of the function of criminal law, 'punishment fits crime' is essentially the unity of retribution function and human right protection function.[61] The retribution function of criminal law requires the punishment of crime, so it inevitably has the tendency to severely punish crime, while the human right protection function requires that the criminal law should not be imposed on the offender too heavy, in order to ensure the fairness and justice of criminal law. Therefore, the balance of two functions leads to the balance of crime and penalty. However, deterrence function has been not added to the doctrine: It is equally important to prevent the occurrence of crime while balancing crime and penalty, and such prevention does not mean the deterrence of potential offenders through extremely harsh penalties, which would lead to the absence

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[60]See, e.g. Markus Dubber and Tatjana Hörnle, *Criminal Law: A Comparative Approach* (Oxford University Press 2014); Chinese criminal law theory has remained largely consistent in these views, see, e.g. Kechang Ma, 'A New Theory of the Function of Criminal Law (刑法的机能新论)' (2009) 8 *People's Procuratorial Semimonthly* (《人民检察》) 10.

[61]Ibid.

of the human right protection function, but rather the suppression of the criminal intensions of potential offenders through a scientific and reasonable design of criminal law.

Economic analysis of criminal law provides a great perspective on this issue. Law and economics generally assumes that offenders are absolutely rational human beings who seek to maximize their own utility in the various acts they commit. Gary Becker's paper in 1968 made a more detailed explanation: A rational offender will only decide to commit a crime under the following circumstances:[62]

$$B > pF \tag{6}$$

in which B represents the benefit, p represents the probability of detection and conviction, and F is the fine (as well as different kinds of punishment). In his view, crime arises because the benefits that can be gained from crime are higher than the costs that it entails, including, but not limited to, the imposition of a prison sentence, moral condemnation, and so on. Subsequent studies have gone a step further and modeled the legitimate wages (w) that potential offenders would normally be able to earn, yielding:

$$B - \mathrm{E}(f|G) > w - \mathrm{E}(f|I) \tag{7}$$

in which benefit of crime minus the penalty for crime will be greater than lawful wage minus the penalty for not committing a crime (i.e., innocence penalty).[63]

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[62]See Gary Becker (n 47).

[63]See, e.g. Robert Cooter and Thomas Ulen, *Law & Economics* (Pearson Addison Wesley 2011) Ch 12 and Ch 13; Orley Ashenfelter and David Card, *Handbook of Labor Economics* (Elsevier 1999) Ch 52.

Based on this, for a potential offender, if the probability of his crime being exposed is p, then the probability of committing a crime but not being punished is $1-p$. And if the benefit of a successful crime is B and the penalty for failure is L, then the utility function of a potential offender can be expressed as:

$$U_i = (1-p)B_i - pL_i \tag{8}$$

which can be further refined for embezzlement and bribery crime: The proceeds of embezzlement and bribery are primarily reflected in the stolen money received, and while other factors such as psychological pleasure are included, they are not primary. Similarly, the costs of embezzlement and bribery are the criminal penalties imposed by the court, which in turn are based on the amount of corruption as the main sentencing criteria.[64] Thus, if we write down the amount involved as m, then the proceeds of crime and the costs of crime, mentioned above, can be further expressed as functions of m. $B(m)$ is a nonlinear function because the marginal utility of money to people is decreasing, so here we retains this function as a whole. $L(m)$ is a function determined primarily by the amount of the crime, mixed with other circumstantial factors, and as shown in Figure 1-2, the sentencing amount standard provides a correspondence from the amount to the length of the sentence, so that there is: $L(m) = k_j m + \varepsilon$, in which k_j means the sentencing amount criteria of j severe level in Tabel 1-2, and ε represents the other factors. On this basis, the aim of the potential offender is to maximize the utility function to give himself the maximum net benefit:

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[64]Taking Chinese legislation as an example, see Appendix I Table 7-8, the amount of embezzlement and bribery is of paramount importance in sentencing and almost directly determines the severity of the sentence.

$$\operatorname{argmax} U_i == \operatorname{argmax}(1-p)B(m_i) - pL(m_i) \Longleftrightarrow \frac{\partial U_i}{\partial m_i} = 0 \qquad (9)$$

and continue the derivation gives:

$$(1-p)\frac{\partial B(m_i)}{\partial m_i} - p\frac{\partial L(m_i)}{\partial m_i} = 0 \Longleftrightarrow (1-p)\frac{\partial B(m_i)}{\partial m_i} = pk_j \qquad (10)$$

Through the above derivation, we can find that a potential perpetrator of embezzlement and bribery will always stop continuing to crime when his marginal benefit (left side) equals his marginal cost (right side). This is because once the marginal benefit is greater, he or she has an incentive to continue to corrupt, while once the marginal cost is greater, it means he or she stops too late and has already suffered a loss. The theory of the crime economics suggests that a potentially rational offender should 'arrange his greed' according to the sentencing amount standard, and once the marginal benefit is greater than the marginal cost, he should do whatever he can to embezzle and bribe, but once the sentencing amount standard is severe and greater than the marginal benefit, he should immediately stop.

In this way, the problems with the legislation prior to the 2016 *Judicial Interpretation* are exposed. The original law divided the sentencing amount standards very unevenly, with the severity increasing very quickly for light and medium crimes, but leveling off immediately at serious crimes, which also leads to the fact that the marginal deterrence of the law decreases rapidly as the amount of corruption and bribery rises This means that someone who embezzles 5,000 Chinese yuan may be sentenced to 1 year in prison, while someone who embezzles 100,000 Chinese yuan will be sentenced to about 10 years, but for those who embezzle 20 million Chinese yuan,

their sentence will probably not exceed 15 years.[65] For a corrupt official who has already embezzled 100,000 Chinese yuan, the cost to him or her of obtaining that 100,000 Chinese yuan would be an average of 10,000 Chinese yuan corresponding to a one-year sentence, but since then, the cost of obtaining another 10,000 Chinese yuan would be far less than a one-year sentence, not even one-month. In this case, the sudden reduction in marginal deterrence will greatly stimulate his or her desire to commit serious crimes and thus keep on embezzling and bribing.

However, this situation is improved with the introduction of the 2016 *Judicial Interpretation*. The steps and tiers of marginal deterrence are more rational designed in *Judicial Interpretation*, and there is no longer a particularly flat sentencing range to encourage potential embezzlers to commit more serious crimes. In addition, the new *Judicial Interpretation* puts further emphasis on the death penalty, which enhances the deterrent effect even more. Offenders must now carefully consider their corrupt decisions, which is the result of the more scientific and reasonable formulation of the *Judicial Interpretation*.

The above theory will lead to the fact that the incidence of severe corruption crimes will decline after the 2016 *Judicial Interpretation*, not only because the penalties are adjusted downward, but also because a more reasonable marginal deterrence system does not leave the soil for the perpetrators of embezzlement and bribery to grab benefits at low cost. This reasoning is similar to the criminal leniency policy that does not easily

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[65]See, e.g. People's Procuratorate of Jiangxi, Ganzhou v Jianyong Hu, Intermediate People's Court of Ganzhou, Jiangxi [2012] Criminal Judgment No 9 (江西省赣州市中级人民法院[2012]赣中刑二初字第 9 号); People's Procuratorate of Zhumadian, Henan v Wei Wu, Intermediate People's Court of Zhumadian, Henan [2013] Criminal Judgment No 20 (河南省驻马店市中级人民法院[2013]驻刑二初字第 20 号), etc.

impose the death penalty, because once stealing 100,000 Chinese yuan will be sentenced to death, killing another person will likewise be a death sentence, and with a marginal cost of zero, this offender will continue to commit vicious crimes indefinitely.

To test whether this mechanism really exists, this paper also conducts an empirical study using a mediating effects test. Based on the data used above, the proportion of cases with sentences greater than 120 months (10 years) is further calculated as a measure of serious cases. Table 5 shows that the interaction term $I_i \times T_t$ does have a significant effect on the serious case ratio, and when the serious case ratio is added to the overall regression model, the coefficients of both it and the interaction term $I_i \times T_t$ are significant, and the coefficient of the interaction term $I_i \times T_t$ is increased, indicating that the addition of the serious case ratio does reduce the effect of the original interaction term $I_i \times T_t$ in the model and the mediating effect exists. In addition, the Sobel test was conducted to prove that the effect does exist significantly.

Table 5. Mechanism analysis

Variables	(9)	(10)	(11)
	Average sentence	Serious case ratio	Average sentence
$I_i \times T_t$	-1.035***	-0.004***	-0.745***
	(-4.11)	(-9.36)	(-4.33)
Serious case ratio			118.257***
			(50.73)
Control variables	Yes	Yes	Yes
City FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Observations	7,159	7,159	7,159
Number of cities	296	296	296
R-squared	0.154	0.060	0.380

Sobel Z	-0.545***
	(-12.34)

Note: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

IV. Did the *Judicial Interpretation* exacerbate the crime?

A. Questions and concerns

Although the above core section has presented a relatively complete empirical study of embezzlement and bribery crime, there is still a need for further discussion because the above discussion of whether *Judicial Interpretation* significantly reduces the average penalty is only one aspect of the question; the other aspect is: Does *Judicial Interpretation* condone the occurrence and reduce the deterrence of embezzlement and bribery crimes? This question is on the opposite side of the discussion of the average penalty, because the two are precisely the result of the deterrence function and human right protection function of criminal law, respectively, and the significant reduction of the average penalty should be worried about whether it is so low that it exacerbates crime and prevents the powerless of the deterrence function caused by the overemphasis on the human right protection function.

In fact, some members of the public have doubts about the introduction of *Judicial Interpretation*. They believe that the decline in average penalties will inevitably lead to the deterrent effect of criminal penalties for embezzlement and bribery being undermined, and that potential offenders will be more likely to go into the abyss of crime under such circumstances, causing more serious damage to society. Moreover, Since China has the tradition of heavy-penaltyism, some people do not feel that the human right protection function of criminal law is more important than the punitive and deterrent functions, and there has been a high demand from the public for severe

punishment for corruption crimes. Therefore, many members of the public believe that the introduction of *Judicial Interpretation* will only promote the occurrence of corruption crimes, and similar adjustments are inappropriate, and some even argue that similar efforts to 'punishment fits crime' are merely excuses for officials to shield themselves. So, what to make of these questions and concerns?

B. Empirical result

It is necessary to give evidence to the above arguments from an empirical perspective based on data from judicial decisions. First, this paper will discuss the question of whether, and then come to the question of why.

According to the theory described above, the introduction of *Judicial Interpretation* that lower penalties will lead to a decrease in the deterrent effect of criminal law, which naturally means that more crimes will be committed. Therefore, this paper first examines the relationship between the introduction of *Judicial Interpretation* and the number of embezzlement and bribery cases in each cities, and the results are shown in the Regreesion (12) in first column of Table 6. From the coefficients of our interaction terms $I_i \times T_t$, the introduction of *Judicial Interpretation* does significantly contribute to the increase in the number of cases, which seems to confirm the public's concern that *Judicial Interpretation* condones crime—but is this really the case?

Table 6. Empirical result

Variables	(12) The number of cases	(13) Total sentence	(14) Light case ratio	(15) The number of cases
$I_i \times T_t$	0.473*** (6.85)	-9.351*** (-3.29)	0.008*** (10.20)	0.471*** (6.88)

Light case ratio				1.070*** (3.30)
Control variables	Yes	Yes	Yes	Yes
City FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	7,159	7,159	7,159	7,159
Number of cities	296	296	296	296
R-squared	0.247	0.148	0.050	0.248
Sobel Z				0.004** (2.11)

Note: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

This paper argues that it is inappropriate to use the number of cases to directly correspond to the social severity of the crime, because such a treatment throws out the important factor of the length of sentence in each case and treats all cases as equivalent, which is very unscientific. In fact, embezzlement and bribery cases show an obvious long-tail distribution, with most cases being relatively minor in severity and a very small number of extremely egregious cases. If the number of cases is used to represent the social harm of the crime, the serious impact of these very few but very bad cases on society is underestimated. Therefore, considering both the number of cases and the severity of the cases, the sum sentences verdicted in each region is a more appropriate measure of the social impact of corruption crimes on a city. The results of this paper are shown in Regression (13) in Table 6, and the results are significantly negative, indicating that the introduction of the *Judicial Interpretation* did not have the effect of amplifying corruption crime in society. To say the least, the argument that the *Judicial Interpretation* condones corruption crime lacks the necessary support.

On this basis, this paper also provides an illustration of the mechanism. As mentioned above, most cases of embezzlement and bribery are minor, and heavy cases are, after all, a minority. If the introduction of the *Judicial Interpretation* has lowered penalties and deterrence, thus encouraging some people to commit crimes, it is unlikely that the crimes and penalties committed by these people will be extremely serious, because the *Judicial Interpretation* features an upward adjustment of the sentencing amount standard, which is more favorable for minor crimes, while the punishment for major crimes is equally severe. Not only that, as mentioned in the design of the mechanism in the previous chapter, the *Judicial Interpretation* introduced a more reasonable division of the sentencing tier, which has the function of reducing serious crimes more. Therefore, this paper argues that although the number of embezzlement and bribery cases has increased to a certain extent because of the introduction of the *Judicial Interpretation*, this increase is mainly caused by the proliferation of minor cases, and the degree of social harm is not particularly high. To address this, this paper constructs the variable of the proportion of sentences less than 36 months as the light case ratio. Using this variable for mediating effect analysis, as shown in the third and fourth columns of Table 6, not only does the introduction of *Judicial Interpretation* significantly increase the proportion of minor cases, the addition of minor cases to the regression does reduce the effect of the interaction term, indicating that this explanatory path does work. As Johann Graf Lambsdorff says, 'restraints and crackdowns are ineffective if they do not provide an emergency exit for those who make minor

mistakes.'[66] Although the number of cases has increased, it is mainly concentrated in minor cases, and this result can ease the public's concern.

C. Discussion and short summary

Finally, this subsection hopes to make a simple abstraction of the above process for better understanding. From the theoretical analysis and empirical results of this paper, the average length of sentence is negatively related to the number of cases, if the penalty is relaxed, the average length of sentence, that is, the punishment for the offender will fall, but affected by the decreasing deterrent effect, more cases will occur; if the penalty rises further, although the deterrent effect of crime rises, the number of cases will fall, but the average length of sentence rises. Once a reasonable limit is exceeded, the fairness to the offender will be lost. This paper proposes that the sum sentence for embezzlement and bribery in a place should be used as a variable to measure the social impact of crime, while the sum sentence is precisely the product of the average sentence length and the number of cases. In this sense, the total sentence can reflect the outcome of the game between the two.

As shown in Figure 5, if the average sentence and the number of cases are strictly inversely related (like $y = kx^{-1}$), then the total sentence, that is, the rectangle shaded in the figure, should always remain constant as the product of the two. For instance, if average penalty falls from P to P_1, then the extra area ($C + D$) and the reduced area (A) should be equal. And from the empirical results of this paper, the introduction of the *Judicial Interpretation* lowers the average sentence and raises the

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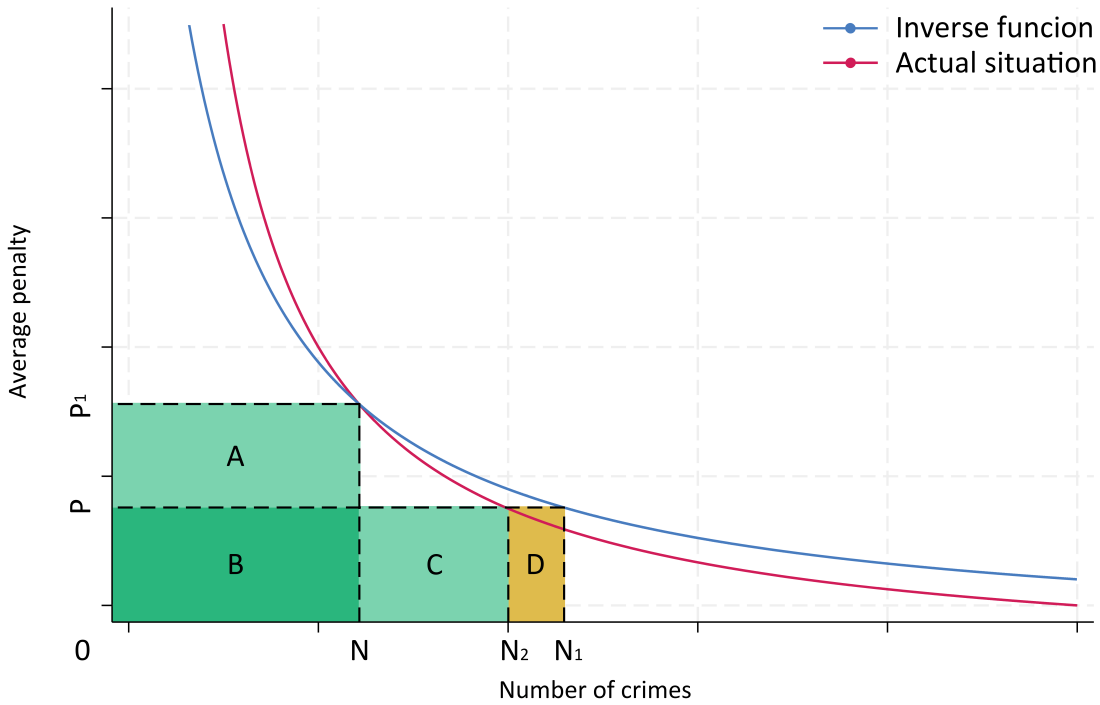
[66]See Johann Garf Lambsdorff, *The institutional Economics of Corruption and Reform: Theory, Evidence and Policy* (Cambridge University Press, 2007) Ch 10.

number of cases, but ultimately lowers the total sum sentence length, which means that the average sentence and the number of cases do not strictly follow the inverse function, but as shown by the red line in the figure. This means that an appropriately lenient sentence for such a crime, although it may lead to a certain degree of increase in the number of crimes, it finally reduce the social harm represented by area D. As China usually adopt the 'pilot first, generalize later' method when implementing major reforms,[67] it is necessary to use judicial data to empirically measure the elasticity of the function between the average penalty and the number of crimes during the pilot period, in order to minimize the total harm to society after the generalization, when implementing similar criminal law reforms that 'punishment fits crime' in the future.

Figure 5. Graph of the social impact of crime



[67]See Ruilong Yang, 'On the Conflict and Harmonization between the Way of Institutional Change and the Goal of Institutional Selection in China (论我国制度变迁方式与制度选择目标的冲突与协调)' (1994) 5 *Economic Research Journal* (《经济研究》) 40.

Now, we can already conclude that Hypothesis 2b is valid, and although *Judicial Interpretation* significantly boosts the number of cases, this does not mean that *Judicial Interpretation* greatly increases the social harm of crime. The above analysis shows that the introduction of *Judicial Interpretation* mainly increased minor crimes, but it also had the effect of reducing the occurrence of vicious crimes, and therefore did not produce particularly serious social consequences.

At this point, the empirical part of this paper has been fully concluded. Hypotheses 1a and 2b hold true, and the *Judicial Interpretation* not only achieves its goal of balancing crime and penalty and promotes the reduction of unreasonable penalties, but also does not seriously condone the occurrence of corruption crimes, and should be given a relatively high evaluation. In response to the question of how to further fine-tune the specific sentencing amount standard, similar theoretical derivations and empirical tests are necessary to further advance in order to develop a deeper understanding.

V. Conclusion

At present, the mitigation of penalties has become the basic guiding principle of criminal legislation and the rational choice of criminal justice in all countries, and has become an important yardstick to measure the degree of social civilization, economic development and human values of a country.[68] The majority of developing countries, including China, are necessary to practice the balance between crime and penalty and the combination of leniency and severity, and realize the unity of order protection function and rights safeguard function of the criminal law, in the process of promoting

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[68]See Bingzhi Zhao and Yixiang Jin (n 5).

the construction of the rule of law in their countries.

The change in sentencing standards for embezzlement and bribery crime in the new 2016 *Judicial Interpretation* is a representative example of the above topic. This paper confirms through empirical research that the introduction of the *Judicial Interpretation* has indeed lowered the average sentence and played a role in promoting 'punishment fits crime', which reveals that even though judicial practice can avoid excessive penalties through flexible discretion in the context of light crime but severe penalty, lowering the sentencing standards through legislative forms can still play a role in lowering penalties and promoting the 'punishment fits crime'. In addition, in the process of adjusting the original overly strict sentencing standards, the scientific and reasonable establishment of a sentencing standard ladder can help reduce the intentions of potential offenders and prevent them from committing more serious crimes because the cost of crime is too low.

On this basis, this paper also responds to the public's question: although the relaxation of sentencing standards has led to an increase in the number of crimes occurring, the overall social harm is decreasing and most of the increase is for minor crimes, which suggests to us that efforts to 'punishment fits crime' do not necessarily condone the crime and increase the social harm. With the goal of reducing the overall social harm and protecting the human rights of the offenders, the balance between the order protection function and the right safeguard function can be scientifically grasped, and the loss suffered by the society due to crime will not be increased. In conclusion, the above conclusions and revelations can serve as a good model for promoting the balance of crime and penalty for similar crimes, in order to further promote the scientificization of China's criminal legislation and advance the construction of the rule of law.

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Appendix I. Detailed changes of the sentencing amount standards before and after the *Judicial Interpretation*

Table 7. Changes of the sentencing monetary amount standards for embezzlement crime and bribery crime before and after the *Judicial Interpretation*

Severity	Before	After	Changes
Light	… Individuals who have engaged in graft with an amount of less than 5,000 Chinese yuan, with the situation being serious, are to be sentenced to less than two years of fixed-term imprisonment or criminal detention …[69]	… Whoever accepts bribes of not less than 10,000 Chinese yuan but less than 30,000 Chinese yuan and has any of the circumstances as prescribed … shall be sentenced to fixed-term imprisonment of not more than three years …[70]	Upward about 5.33 times
	… Individuals who have engaged in graft with an amount of more than 5,000 Chinese yuan but less than 50,000 Chinese yuan are to be sentenced to more	… Whoever embezzles or accepts bribes of not less than 30,000 Chinese yuan but less than 200,000 Chinese yuan … shall be sentenced to fixed-term imprisonment of not more than three years or criminal detention …[72]	Upward about 11.15 times

[69] *Criminal Law 1997* (n 13) Art 383(4).

[70] *Judicial Interpretation* (n 7) Art 1(1-2).

[72] *Judicial Interpretation* (n 7) Art 1(1).

	than one year but less than seven years of fixed-term imprisonment …[71]		
Medium	… Individuals who have engaged in graft with an amount of more than 50,000 Chinese yuan but less than 100,000 Chinese yuan are to be sentenced to more than 5 years of fixed-term imprisonment …[73]	… Whoever embezzles or accepts bribes of not less than 200,000 Chinese yuan but less than 3 million Chinese yuan … shall be sentenced to fixed-term imprisonment of not less than three years but not more than ten years …[74]	Upward about 24.61 times
Severe	… Individuals who have engaged in graft with an amount of more than 100,000 Chinese yuan are to be sentenced to more than 10 years of fixed-term imprisonment or life imprisonment …[75]	… Whoever embezzles or accepts bribes of not less than 3 million Chinese yuan … shall be sentenced to fixed-term imprisonment of not less than ten years, life imprisonment, or death …[76]	Upward about 30 times
Average (weighted by 60% for light, 30% for medium and 10% for heavy, based on real-world crime structure)			Upward about 15.33 times

[71] *Criminal Law 1997* (n 13) Art 383(3).

[73] *Criminal Law 1997* (n 13) Art 383(2).

[74] *Judicial Interpretation* (n 7) Art 2(1).

[75] *Criminal Law 1997* (n 13) Art 383(1).

[76] *Judicial Interpretation* (n 7) Art 3(1).

Tabel 8. Changes of the sentencing monetary amount standards for misappropriation of public funds crime before and after the *Judicial Interpretation*

Severity	Before	After	Changes
Light	… Where a person misappropriates public funds for personal use and for conducting profit-making activities… 10,000 to 30,000 Chinese yuan shall be the starting point of 'relatively large' …[77]	…Where a person misappropriates public funds for personal use and for conducting profit-making activities … if the amount involved is not less than 50,000 Chinese yuan, it shall be determined that 'the amount is relatively large' …[78]	Upward about 2.5 times
	… Whoever misappropriates public funds for personal use and for conducting illegal activities, 5,000 to 10,000 Chinese yuan shall be the starting point of 'relatively large' …[79]	… Whoever misappropriates public funds for personal use and for conducting illegal activities … shall be subject to criminal liability … if the amount involved is not less than 30,000 Chinese yuan …[80]	Upward about 4 times

[77]Interpretation of Several Issues concerning the Application of Law in the Handling of Criminal Cases of Misappropriation (关于审理挪用公款案件具体应用法律若干问题的解释) (hereinafter referred to as the *Judicial Interpretation of Misappropriation*) Art 3(1).

[78]*Judicial Interpretation* (n 7) Art 6.

[79]*Judicial Interpretation of Misappropriation* (n 77) Art 3(1).

[80]*Judicial Interpretation* (n 7) Art 5.

Medium	… Whoever misappropriates public funds for personal use and for conducting illegal activities, more than 50,000 to 100,000 Chinese yuan shall be one of the situation of 'serious circumstance' …[81]	… Any of the following circumstances shall be determined as 'serious circumstance': … (1) The amount of misappropriated public funds is not less than 1 million Chinese yuan …[82]	Upward about 13.33 times
Severe	… The starting point for 'the amount is large' is 150,000 to 200,000 Chinese yuan …[83]	… If the amount involved is not less than 5 million Chinese yuan, it shall be determined that 'the amount is large' …[84]	Upward about 28.57 times
Average (weighted by 60% for light, 30% for medium and 10% for heavy, based on real-world crime structure)			Upward about 8.81 times

[81] *Judicial Interpretation of Misappropriation* (n 77) Art 3(2).

[82] *Judicial Interpretation* (n 7) Art 5.

[83] *Judicial Interpretation of Misappropriation* (n 77) Art 3(1).

[84] *Judicial Interpretation* (n 7) Art 6.